

Diffusion-Based Data Augmentation for Imbalanced Medical Image Classification

MSML612 Group Project, Interim Report

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Abstract

Clinical image datasets are long-tailed: the diagnoses that matter most are often the rarest, and classifiers trained on such data underperform precisely where reliable detection is most valuable. We ask whether diffusion-generated images can improve a downstream classifier on rare classes, and whether they beat classical and GAN augmentation in a controlled, leakage-free comparison. Our testbed is HAM10000, a severely imbalanced dermoscopic benchmark. To date we have built and verified the full data pipeline, whose lesion-level stratified split eliminates a substantial test-set leakage present in naive image-level splitting while retaining every image, and we have run a feasibility study that fixed our generative architecture, a class-conditional 37.1M-parameter pixel-space DDPM trained from scratch. Remaining work is model training and the four-way augmentation comparison, for which the measurement protocol is fully specified below.

1. Problem Statement and Research Question

Deep networks match expert accuracy on many medical imaging tasks, but only on large, balanced datasets, a condition that rarely holds clinically. The standard remedy is data augmentation. Classical augmentation (rotations, flips, color jitter) only perturbs existing pixels and adds no new anatomical variation. GANs can synthesize new samples but suffer unstable training and mode collapse on limited data. Diffusion models are the current state of the art in image generation: high fidelity, diverse samples, stable training, and native class-conditional control. This motivates our central research question.

Can diffusion-generated synthetic images meaningfully improve a downstream medical classifier on rare classes, and does diffusion augmentation outperform classical and GAN-based augmentation?

The course requires choosing between a transformer and a diffusion model. That choice is settled by the task: ours is image generation, for which diffusion is the natural, state-of-the-art choice.

2. Data Preparation and Curation

2.1 Dataset

HAM10000 (Tschandl et al., 2018) contains 10,015 dermoscopic images across 7 diagnostic classes, publicly available from the Harvard Dataverse (doi:10.7910/DVN/DBW86T). It is a well-studied, severely imbalanced benchmark, which makes it an ideal testbed for a rare-class augmentation study. The imbalance is stark: melanocytic nevi (nv) accounts for 6,705 images (66.9%), roughly 58 times the 115 images of dermatofibroma (df). The per-class counts appear in the split table in Section 2.3; the rare classes there are the ones augmentation must help.

2.2 The leakage problem, quantified

HAM10000 provides a `lesion_id` for every image, and many lesions are photographed more than once. In the metadata, 4,501 of 10,015 images (44.9%) share a lesion with at least one other image; 1,956 lesions have 2 or more images (up to 6). Two images of the same lesion are near-duplicates.

A naive image-level random split therefore leaks. Averaged over 200 random image-level 70/15/15 splits (`preprocessing/leakage_analysis.py`), 34.9% of test images share a lesion with a training image (std 1.1). Critically, the leakage is worst on exactly the minority classes the study depends on: mel 62.1%, df 51.1%, bcc 50.2%, bkl 46.4%, vasc 42.6% and akiec 41.9%, against only 26.6% on the majority class nv. A model evaluated under such a split would report inflated numbers driven by memorized duplicates, not generalization, which directly threatens the study's core claim: a measured F1 delta on the test set.

2.3 Our split: lesion-level, stratified, all images kept

We split at the lesion level rather than the image level: every image of a given lesion is assigned to the same split. Splitting is stratified by class so each split preserves the class distribution, using a fixed seed (612) and a 70/15/15 ratio. A runtime assertion verifies that no `lesion_id` appears in more than one split; the split passes with zero crossing lesions.

We depart from the proposal deliberately here. It planned to de-duplicate by lesion; we instead keep every image and move the boundary to the lesion. That yields 10,015 training-eligible images versus 7,470 unique lesions, a 34% increase in usable data at no leakage cost. For rare classes with only tens of training images, discarding a valid second view is a real loss we avoid.

The resulting splits, with the full class distribution (sorted by frequency):

Class	Full name	Train	Val	Test	Total	Share
nv	Melanocytic nevi	4,679	1,015	1,011	6,705	66.9%
mel	Melanoma	776	163	174	1,113	11.1%
bkl	Benign keratosis	765	165	169	1,099	11.0%
bcc	Basal cell carcinoma	354	77	83	514	5.1%
akiec	Actinic keratosis / Bowen's disease	227	55	45	327	3.3%
vasc	Vascular lesions	96	23	23	142	1.4%
df	Dermatofibroma	84	15	16	115	1.1%
All		6,981	1,513	1,521	10,015	100%

The split is written once to `data/splits.csv` (image_id, lesion_id, dx, split) and is the single source of truth for every downstream consumer. df, with 84 training images, is the hardest case and the main stress test for the generator.

2.4 Materialization and normalization

Images are sorted into `data/labeled/<class>/` and the split is materialized into `data/splits/{train,val,test}/<class>/` directory trees using hard links, which is near-free on disk. This lets standard loaders (PyTorch `ImageFolder`, Keras `image_dataset_from_directory`, StyleGAN2-ADA) read a leak-free tree directly, instead of relying on their built-in validation-split arguments, which shuffle within a class and would re-introduce the exact lesion leakage we removed. A shared `HAM10000` dataset class provides fixed global class indices and two normalization modes: `[-1, 1]` for the generative model and ImageNet statistics for the classifier.

3. Neural Network Design

The project has three model components: the diffusion generator (ours, the core contribution), a fixed downstream classifier (the measurement instrument), and two augmentation baselines.

3.1 Generative model: class-conditional pixel-space DDPM

We implement a class-conditional Denoising Diffusion Probabilistic Model (Ho et al., 2020), trained from scratch in pixel space at 64x64. We adapt the Hugging Face diffusers `UNet2DModel` and `DDPMScheduler` (von Platen et al., 2022) from random initialization; the adaptation is the class-conditioning embedding, the classifier-free-guidance dropout, the 64x64 channel configuration, and the domain training, while the residual-block and attention machinery is reused from the library. The architecture is concrete:

Field	Value
Base implementation	diffusers UNet2DModel + DDPM Scheduler , random init
Input	64 x 64 x 3
Channels per stage (64, 32, 16, 8)	128, 256, 256, 256
Residual blocks per stage	2
Self-attention	at the 16x16 and 8x8 stages
Class conditioning	8 embeddings (7 classes + 1 null for guidance), added to the timestep embedding
Parameters	37.1M

Training configuration (planned): AdamW at learning rate 1e-4, batch size 64, up to roughly 100k steps, exponential moving average of weights (decay 0.9999). The forward process uses a cosine noise schedule (Nichol and Dhariwal, 2021) with $T = 1000$ steps; sampling uses DDIM at 50 steps. Classifier-free guidance (Ho and Salimans, 2021) drops the class label to the null token with probability 0.1 during training, and at inference we sweep the guidance weight over $\{1.0, 2.0, 3.0\}$ to trade diversity against class fidelity, which is the control minority-class augmentation needs. At 37.1M parameters and 64x64 the model fits a single 12 GB GPU with room to spare, at an estimated single-digit GPU-hours to convergence, with the UMD Zaratan cluster available for the multi-seed sweeps.

Two design choices carry most of the argument:

- **One shared model, not one per class.** A single conditional model sees all 6,981 training images and reuses representations across classes, while the class embedding steers output. This is our primary defense against memorization on df (84 images), where a dedicated per-class model would almost certainly overfit. To ensure the rare classes receive enough gradient exposure to benefit, DDPM training uses class-balanced sampling, matching the classifier's policy (Section 3.2), rather than the natural 67:1 frequency.
- **From-scratch, pixel-space, at 64x64.** We evaluated the latent-diffusion route first and set it aside on the evidence in Section 5.1. We also depart from the proposal's 256x256 target and generate at 64x64, which keeps the from-scratch DDPM, the StyleGAN2-ADA baseline, and the multi-seed comparison all tractable on our compute budget.

3.2 Downstream classifier (measurement instrument)

The augmentation question is answered by a classifier held identical across all four regimes: a ResNet-18 trained from scratch at 64x64 with class-balanced sampling. Random initialization keeps the whole pipeline free of pretrained weights and is well matched to the 64x64 input, where an ImageNet-pretrained backbone expecting 224x224 gains little. Its architecture, optimizer, schedule, and seeds are fixed across regimes; only the training-set composition changes, so the augmentation strategy is the only variable.

3.3 The four regimes

All four regimes share the same classifier, the same 64x64 resolution, and the same real-only validation and test sets. No synthetic image ever touches evaluation.

1. **Real data only** (class-balanced sampling): the reference condition. Note it already oversamples minority images through balanced sampling, so it is a fair floor rather than a pure no-op.
2. **Classical augmentation**: horizontal and vertical flips, rotations, and color jitter, as implemented in `src/dataset.py`.
3. **GAN augmentation**: class-conditional StyleGAN2-ADA (Karras et al., 2020), the standard limited-data GAN, run with its published default recipe so it is not under-tuned relative to our diffusion model.
4. **Diffusion augmentation (ours)**: synthetic minority-class images from Section 3.1.

Regimes 2 to 4 mix synthetic or augmented images into the real training set; the classifier is retrained and evaluated on the untouched real test set. To keep the comparison honest, all four regimes are trained for the same number of gradient steps over the same total image count, and we add a replication control that oversamples real minority images to match the synthetic regimes' data volume without new content, isolating gains from volume alone versus synthetic diversity.

The full pipeline:

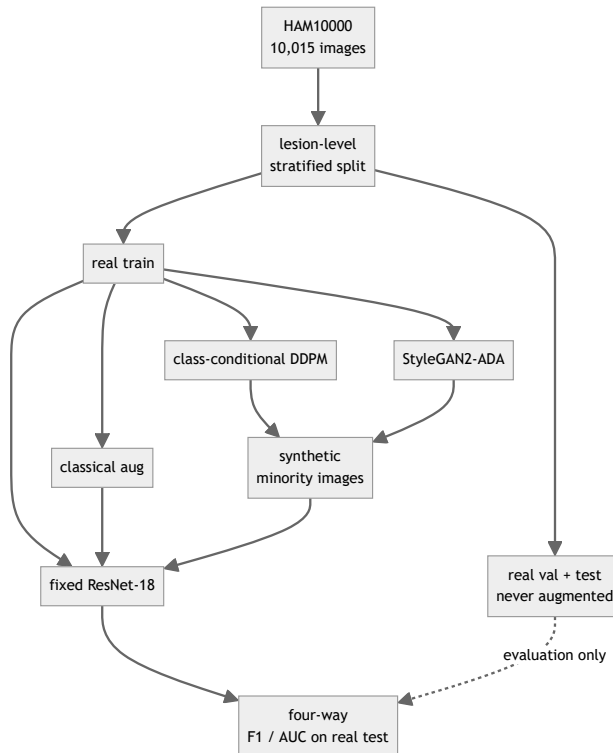


Figure 1. Experimental pipeline. Synthetic and augmented data enter only the training set; validation and test stay real-only.

4. Implementation and Code

The repository is organized for reproducibility with no manual steps between raw data and a leak-free, materialized split.

```
preprocessing/  
  sort_images.py      # raw images -> data/labeled/<class>/ (copy, keep all 10,015)  
  make_splits.py      # lesion-level stratified split -> data/splits.csv (seed 612)  
  make_split_dirs.py  # splits.csv -> data/splits/{train,val,test}/<class>/ (hard links)  
  leakage_analysis.py # quantifies naive-split leakage vs our lesion-level split  
src/  
  dataset.py          # shared HAM10000 loader, fixed class indices, 2 normalizations  
  vae_roundtrip.py    # autoencoder feasibility study (Section 5.1)  
requirements.txt      # pinned deps, CUDA 12.6 index
```

Two properties make the pipeline reproducible without manual intervention. For verified integrity, `make_splits.py` asserts at runtime that no lesion crosses splits, so a leaking split fails loudly instead of silently, and `leakage_analysis.py` regenerates the Section 2.2 numbers on demand. For a pinned environment, `requirements.txt` fixes exact versions against a CUDA 12.6 wheel index, and the fixed split seed (612) reproduces `data/splits.csv` byte for byte.

Implemented and runnable today: the four preprocessing scripts, the shared dataset loader, and the feasibility study. In progress: the DDPM training and sampling scripts, and the classifier and evaluation harness.

5. Progress, Results, and Evaluation Protocol

5.1 Result: autoencoder feasibility study

Before committing to an architecture we asked whether the image content survives compression, which sets an upper bound on any latent approach. Encoding and decoding held-out images through the pretrained SD autoencoder gives a mean round-trip PSNR of 35.1 dB, consistent across classes (32.6 to 38.0 dB), with pigment networks, lesion borders, and hairs preserved (Figure 2). Two conclusions: latent diffusion would not be bottlenecked by the decoder, and, combined with the project's from-scratch requirement (training our own autoencoder as well is out of scope for the timeline), this justified building a pixel-space model directly rather than depending on pretrained latents.

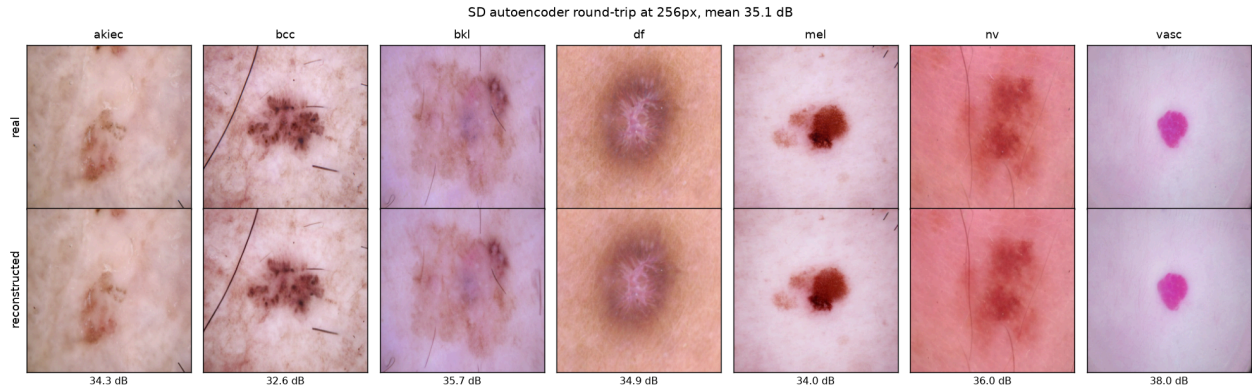


Figure 2. Real (top) vs autoencoder reconstruction (bottom), one per class, with per-class PSNR. Deliberately run at the proposal's original 256px to bound what a latent approach could have achieved, which is why its resolution differs from the final 64x64 model.

5.2 Result: data integrity

The delivered split has zero lesions crossing the train/val/test boundary (asserted), against a 34.9% test-set leak rate under the naive image-level baseline (Section 2.2). This is the foundation that makes any later F1 delta trustworthy.

5.3 Evaluation protocol (specified, pending execution)

The downstream augmentation results are the project's main outcome and are not yet available, since model training is the next phase. The immediate next measurable milestone is the real-data-only classifier baseline. The measurement protocol is fixed:

- **Primary metric:** per-class F1 on the rare classes, reported as a delta over the real-data-only baseline. Success is diffusion augmentation (regime 4) beating regimes 1 to 3.
- **Secondary metrics:** macro-averaged F1, balanced accuracy, and per-class AUC.
- **Significance:** each regime is trained with 5 seeds, paired on seed (identical initialization and data order, only the injected images differ). We test per-class F1 deltas with a paired Wilcoxon signed-rank test and apply Holm-Bonferroni correction across the 7 classes. Because the rare-class test sets are tiny (df 16, vasc 23, akiec 45 images, so F1 moves in coarse steps), we supplement seed variance with bootstrap confidence intervals resampled from the test-set predictions.
- **Ratio selection:** the synthetic-to-real mixing ratio for each regime's headline number is chosen on the validation set; the test set is evaluated once per regime.
- **Generative quality:** FID (Heusel et al., 2017) is reported per class, not pooled (a pooled score is dominated by nv), and as a relative within-study ranking only, since at 64x64 it is not comparable to literature values benchmarked at higher resolution. We add Kernel Inception Distance (KID), which is less biased on the small rare-class reference sets (df, vasc, akiec).
- **Memorization check:** every generated image gets a nearest-neighbor distance to the real training set of its class (LPIPS and Inception features), compared against the real-to-real nearest-neighbor distribution within that class. Samples below the 5th percentile are flagged as suspected copies and discarded before they reach the classifier; we report the flagged fraction per class.

- **Ablations:** a mixing-ratio sweep (0, 25, 50, 100% synthetic added to the real minority set) and a generator sample-size study (DDPM trained on 25, 50, 100% of the real rare-class images), to locate where synthetic data helps most.

Regime	Rare-class F1	Macro F1	FID (per class)	Status
Real data only	pending	pending	n/a	protocol fixed
Classical	pending	pending	n/a	protocol fixed
GAN (StyleGAN2-ADA)	pending	pending	pending	protocol fixed
Diffusion (ours)	pending	pending	pending	protocol fixed

Conclusions from this study are established at 64x64; resolution-scaling behavior toward full-resolution dermoscopy is out of scope. If a from-scratch DDPM fails to produce legible rare-class samples, the fallbacks are heavier EMA and regularization, additional class-balanced oversampling, or narrowing scope to the classes that do converge.

6. Related Work

Diffusion foundations. DDPM (Ho et al., 2020) established the denoising formulation we build on; Improved DDPM (Nichol and Dhariwal, 2021) contributed the cosine noise schedule we adopt; latent diffusion (Rombach et al., 2022) moved generation into a compressed latent space, the route our feasibility study evaluated. Dhariwal and Nichol (2021) showed diffusion surpasses GANs on image synthesis, and classifier-free guidance (Ho and Salimans, 2021) provides the conditional control our minority-class task requires.

Generative augmentation. Frid-Adar et al. (2018) showed GAN-synthesized medical images improve a CNN on liver-lesion classification, an early positive result for synthetic augmentation. Trabucco et al. (2024) demonstrated diffusion-based augmentation gains on general vision benchmarks. Whether such gains are real information or re-encoded generator bias is still debated, which our controlled, leakage-free comparison is designed to address.

Baselines and metrics. StyleGAN2-ADA (Karras et al., 2020) is the standard limited-data GAN and our GAN baseline; FID (Heusel et al., 2017) is the standard generative-quality metric. HAM10000 (Tschandl et al., 2018) is our dataset, and we adapt the diffusers library (von Platen et al., 2022) for the diffusion model.

7. Remaining Work and Timeline

Phase	Status
Data pipeline and leak-free splits	Done
Architecture feasibility study	Done
DDPM training and sampling	Next
Classifier and evaluation harness	In progress
Classical and StyleGAN2-ADA baselines	Planned
Four-way comparison, ablations, significance tests	Planned
Final figures, report, presentation	Planned

Training runs use local GPUs and the UMD Zaratan HPC cluster (Slurm, mixed precision) as needed. The nearest milestones are the real-data-only classifier baseline and a trained conditional DDPM producing recognizable per-class samples, followed by the first no-augmentation-vs-diffusion comparison.

8. References

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